REG.NO.:

VIT
Vellore Institute of Technology

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT:B1+TB1

Programme Name & Branch	:	**MCA**
Course Code and Course Name	:	**PMCA507L - Machine Learning**
Faculty Name(s)	:	**Dr. A. Anitha and Dr. M. Parimala**
Class Number(s)	:	**3327 and 3331**
Date of Examination	:	**28th Jan 2025**
Exam Duration	:	**90 minutes**

Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes (Type the CO statements covered in this question paper. Use the CO number as per the syllabus copy)
  CO1 - Recognize the characteristics of machine learning that makes it useful to solve real-world problems
  CO2 - Provides solution for classification, regression and clustering approaches in real-world applications

Q. No	Question	M	CO	BL
1.	Illustrate with an example of your understanding about the concept of version space.	10	CO1	BL2
2.	Draw the multilayer perceptron based on the given information and find the output using the activation functions as mentioned below: **Input vector** : $[x_1, x_2, x_3] = [1.0 \quad 2.0 \quad 3.0]$ **Weights** : (i) Input to hidden Layer (h1 and h2) are $wh_1 = [0.1, 0.2, 0.3]$ *and* $wh_2 = [0.4, 0.5, 0.6]$ (ii) Hidden to Output Layer: $w_0 = [0.7, 0.8]$ **Biases** : (i) Hidden Layer : $bh_1 = 0.1$ *and* $bh_2 = 0.2$ (ii) Output Layer : $b_0 = 0.3$ **Activation Functions:** (i) Input to Hidden layer: ReLU Activation function (ii) Hidden Layer to Output layer: Binary Sigmoid function	10	CO2	BL6
3.	Find the maximally specific hypothesis for the given training dataset. Some of the abbreviations as given as follows: JAPAN as JAP, USA as US, HONDA as HO, TOYOTA as TO, CHRYSLER as CH, Economy as ECO , Sports car as SPO.	10	CO1	BL4

REG.NO.:

VIT
Vellore Institute of Technology

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
CONTINUOUS ASSESSMENT TEST - I
WINTER SEMESTER 2024-2025

SLOT:B1+TB1

Instances	Country	Manuf.	Color	Year	Type	Class
X1	JAP	HO	Blue	1980	ECO	+ve
X2	JAP	TO	Green	1970	SPO	-ve
X3	JAP	TO	Blue	1970	ECO	+ve
X4	US	CH	Red	1980	ECO	-ve
X5	JAP	HO	White	1980	ECO	+ve
X6	JAP	TO	Green	1980	ECO	+ve
X7	JAP	HO	Red	1990	ECO	-ve

4. What is regression? Form the regression line of equation and predict the price for 25 sq.ft.

Also evaluate using any of the **two measures**

Sq.Ft(X)	Price(Y)
10	240
12	220
15	300
18	340
20	420

10 CO2 BL5

5. Predict the crop yield based on the given dataset for the features as {Rainfall = Medium and Temperature = High} using Naive Bayes classifier.

Rainfall	Temperature	Crop Yield (Class)
Low	Low	Low Yield
Low	High	Medium Yield
Medium	High	High Yield
High	High	High Yield
High	Low	Low Yield
Medium	Low	Medium Yield
Low	High	Low Yield
High	High	High Yield

10 CO2 BL6

****